

# MALMI WITHANAGE

Galle, Sri Lanka (Open to Remote - GMT+5:30) | malmiwithanage@gmail.com | +94 76 230 9951

[github.com/malmiwithanage](https://github.com/malmiwithanage) | [linkedin.com/in/malmi-withanage](https://linkedin.com/in/malmi-withanage) | [malmi-dev.vercel.app](https://malmi-dev.vercel.app)

Associate Software Engineer specializing in robust backend systems, secure multi-tenant isolation, and data pipelines. Experienced in building event-driven worker layers and shipping production-grade, peer-reviewed machine learning applications. Committed to rigorous backend test coverage, data consistency, and end-to-end service deployment.

## TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, Python, Java, C++, HTML5, CSS3
- **Frontend:** React, Next.js, React Native (Expo), Tailwind CSS, State Management
- **Backend Architecture:** NestJS, FastAPI, Node.js, REST APIs, Microservices, JWT, OAuth 2.0, RabbitMQ
- **Data & Machine Learning:** PostgreSQL, MongoDB, Redis, Scikit-learn, XGBoost, Random Forest, SARIMA, Pandas, NumPy
- **Cloud & DevOps:** AWS (EC2, RDS), Microsoft Azure, Docker, Cloudflare, Git, GitLab CI/CD, BullMQ

## PROFESSIONAL EXPERIENCE

**Associate Software Engineer** | Lanka Communication Services (Pvt) Ltd Oct 2025 – Present

- Architected and expanded core features for an enterprise Learning Management System (LMS) using NestJS, Next.js, RabbitMQ, and MongoDB to meet strict delivery timelines.
- Designed and shipped a complex timetable scheduling module, drastically reducing post-deployment bugs and hotfixes through rigorous local testing with Docker and Redis cache pipelines.
- Mentored engineering interns, contributed in daily Agile standups, and drove cross-project issue resolution to stabilize legacy codebases.

**Trainee Associate Software Engineer** | Lanka Communication Services (Pvt) Ltd Jul 2025 – Oct 2025

- Accelerated full-stack feature development for a high-traffic LMS platform, coordinating closely within an Agile team to consistently hit critical deployment deadlines.

**Junior Software Engineer** | 2xDev Mar 2025 – Jun 2025

- Owned the resolution of high-priority web application bugs and actively contributed to decision-making processes for scaling core digital products.
- Engaged in rapid Agile delivery frameworks and daily engineering standups to coordinate hotfixes and feature releases alongside cross-functional team members.

**Software Engineer Intern** | 2xDev Jun 2024 – Dec 2024

- Developed scalable REST API endpoints and interactive user interface components using modern JavaScript/TypeScript and web frameworks.

## TECHNICAL PROJECTS

**Logi Track — Multi-Tenant Logistics System** (NestJS, PostgreSQL, Prisma, Redis, BullMQ, Supabase)

- Enforced absolute tenant data isolation utilizing a custom cryptographic JWT Guard layer, preventing access parameter tampering.
- Engineered an asynchronous CSV export pipeline using BullMQ and Redis, offloading heavy computations from the synchronous request thread.
- Implemented automated data isolation verification using Jest integration suites, covering complex cross-tenant rejection tests.

**CinoGrow — ML Agriculture Decision Support System** (FastAPI, Python, XGBoost, SARIMA, React Native, Docker, AWS)

- Built dynamic distillation time regression models ( $R^2$  0.970) and oil-quality classification loops using 5-fold cross-validation.
- Designed a SARIMA time-series model (3.85% MAPE) to predict volatile seasonal price fluctuations for smallholder farmers.
- Containerized the entire application stack via Docker, enabling seamless automated deployment to production clusters on AWS.

## ACADEMIC PUBLICATION

**18th ECAI 2026 International Conference on Electronics, Computers and Artificial Intelligence (Bucharest, Romania)**

"CinoGrow: An Integrated Machine Learning Decision Support System for Precision Cinnamon Farming" Co-authored with SLIIT research cohort. Supervised by Ms. S. Silva & Ms. N. Gamage. DOI: 10.1109/ECAI69016.2026.11613746

## EDUCATION

**BSc (Hons) in Information Technology** | Sri Lanka Institute of Information Technology (SLIIT)

2022 – 2026